MIL-C-22016(AS)
Amendment 3
26 September 1966
Superseding
Amendment 2
3 December 1964

MILITARY SPECIFICATION

CARTRIDGES (STANDARD), MARTIN-BAKER MARK 5,
EJECTION SEAT

This amendment forms a part of Military Specification MIL-C-22016(NOrd) dated 7 May 1959 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment basic specification MIL-C-22016(NOrd) is changed to read MIL-C-22016(AS).

Page 1, in block, change "FSC 1375" to read "FSC 1377".

Page 5, paragraph 3.6.1 - Change to read, "Catapult cartridges shall function properly when subjected to the test specified in 4.4.9.1 and/or 4.4.10. Procuring activity shall specify whether tests of 4.4.9.1 or 4.4.10 or both shall apply."

Page 5, paragraph 3.6.2 - Change to read "Drogue gun cartridges shall function properly when subjected to the test specified in 4.4.9.2 and/or 4.4.10.1. Procuring activity shall specify whether tests of 4.4.9.1 or 4.4.10 or both shall apply."

Page 7, paragraph 4.3.3 - Under the heading of test, add the following after 4.4.9 "and 4.4.10*."

Page 11, paragraph 4.4.10 and 4.4.10.1 - Change paragraph "4.4.10 Retest etc., - and 4.4.10.1 Test Failure etc.," to read as follows:

"4.4.10 Ballistic tests. The government agency designated by the procuring activity shall fire thirty sets of catapult cartridges (sample size in accordance with Table B-1, Code Letter J, AQL .04, MIL-STD-414) in a Martin-Baker Catapult mounted on a car weighing 364 pounds, riding on a horizontal track. In addition to the sample of catapult cartridge sets required for the above test, three additional catapult cartridge sets shall be furnished from the production lot for investigative purposes. Friction loss in the system shall not exceed five percent. All cartridges shall be conditioned at a temperature of 70°F ± 5°F for a period of not less than 24 hours

*Test to be as prescribed by the procuring activity.

FSC 1377

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prior to test firing. Cartridge temperature for each firing shall be 70°F ± 5°F. A record of pressure versus time shall be taken on each firing. Velocity shall be measured over a six inch interval (from tube separation to six inches past) with suitable instrumentation. Acceptability shall be determined in accordance with the following criteria:

Maximum Pressure:	The upper limit (U) of the maximum pressure shall be 1620 psi (corresponds to 23 g).
Rate of Pressure Rise:	The upper limit (U) of the rate of pressure rise shall be 24,800 psi/sec. (corresponds to 350 g/sec.).
Velocity:	The lower limit (L) of the velocity shall be 64.0 ft/sec.

The quantities $(U - \bar{X})/S$ or $(\bar{X} - L)/S$, whichever applies, shall be equal to or greater than the acceptability constant K = 2.73 as taken from Table B-1 of MIL-STD-414, AQL = .04.

4.4.10.1 Drogue gun cartridges. Government agency designated by the procuring activity shall fire 30 drogue gun cartridges under conditions specified in 4.4.9.2. Acceptability is to be based in accordance with the following:

Delay Time:	The upper limit (U) of the delay time from impact of firing pin to maximum pressure shall be 0.05 second.

The quantity $(U - \bar{X})/S$ shall be equal to or greater than the acceptability constant K = 2.73 as taken from Table B-1 of MIL-STD-414, AQL = .04.

4.4.10.2 Hangfires or Misfires. One or more hangfires or misfires of catapult or drogue gun cartridges shall cause rejection of lot represented as specified in 4.4.9.3 and 4.4.9.4, respectively."

Page 11, paragraph 4.4.11 - Delete paragraph 4.4.11 - Extremes of Temperature, etc. and substitute: "4.4.11 Retest. There shall be no retests."

"4.4.11.1 Test failure. If test failure is due to an assignable cause, excluding test cartridges, original test results shall be discarded and the test reconducted with a new sample."

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Page 11, following paragraph 4.4.11.1, add the following paragraph: "4.4.12 Extremes of temperature. Whenever changes occur in propellant lot in the Martin-Baker MARK 5 Ejection Seat Catapult and Drogue Gun Cartridges, ballistic tests shall be conducted on 60 sets of cartridges by the activity designated by the procuring agency. Cartridges shall be fired over the temperature range from -50°F to +160°F (±5°F), in order to verify the validity of charge determination."

Page 11, paragraph 4.5 - Delete in its entirety and substitute: "4.5 Test results. Results of the static firing (4.4.9) and ballistic tests (4.4.10) shall be forwarded to the Naval Air Systems Command."

Pages 12, 13, 14, and 15, delete paragraphs 6 through 6.6.8 in their entirety and insert in lieu thereof the following:

"6. Explosives safety precautions.

6.1 Applicability. The loading, assembly and handling of the explosives, subassemblies and the finished items covered by this specification, involve hazardous operations and therefore require suitable explosives safety precautions. At plants of the Naval Air Systems Command, Department of the Navy, and other Government plants where the Government is responsible for safety, the precautions listed below are mandatory. These same minimum provisions listed below are recommended for observance by a contractor or manufacturer that explosively loads the item covered in this specification in order to fulfill his own safety responsibility. Other warnings and precautions, pertaining to the operational effectiveness or safety of the specified item during its use (for example, during its handling, launching or firing) are included in the detailed technical requirements of the specification. Use of this specification shall not be construed to replace other safety regulations of governmental agencies or to relieve the contractor or manufacturer of responsibility for the safety of his operations.

6.2 Personnel and equipment. Proper care must be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In connection with the above, the precautionary measures in the following paragraphs should be observed. (See also paragraph 6.4, Static and electrical and other applicable parts of these safety precautions.)

6.2.1 Explosives operations shall be kept under strict supervision. Periodic inspections shall be employed to assure compliance with established safety regulations.

6.2.2 Safe equipment and methods should be utilized for handling and transporting explosives and loaded parts.

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6.2.3 Where required, remote control barricaded handling equipment shall be used for explosives operations such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.2.4 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.2.5 Maintain equipment such as presses, dies, crimpers, blenders, mixers, sifters, conveyors, and the like, in safe operating condition.

6.2.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

6.2.7 During static tests, propulsion units such as rocket motors shall be restrained against full thrust to prevent flight if ignited, for example, during test of igniter circuit. Also, means to vent flames and exhaust gases harmlessly shall be provided.

6.3 Operating procedures. All loading operations shall be conducted in full accordance with a formalized operating procedure which shall include the following:

6.3.1 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operations.

6.3.2 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.3.3 Appropriate safety instructions for specific operations shall be posted and personnel instructed as to the hazards present and precautions involved.

6.3.4 The number of working personnel and transients permitted in each work space shall be established and posted in such areas. The numbers shall be the minimum number required to accomplish necessary operations.

6.3.5 All loading operations should be conducted in a neat and orderly manner.

6.3.6 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings.

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6.3.7 Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from, floors, walls, ceiling, ledges, tables, benches, piping, and equipment or the items loaded. Clean up any spilled material immediately.

6.3.8 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.3.9 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which affect functioning should, insofar as practicable, avoid using bare fingers or improper equipment, in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.3.10 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges. (See paragraph 6.3.4).

6.4 Static and electrical.

6.4.1 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges shall be employed. Fabrics such as silk and nylon, which promote static generation shall be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low energy type electric primers, detonators and squibs. The latter types of items shall have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding from induced electric currents originated by sources such as lightning, static, electromagnetic radiations, etc. Where necessary for safety, humidity of work rooms should be appropriately increased, as required to lessen electrostatic effects but without inducing excessive moisture absorption by any of the components of the item being processed.

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6.4.2 Protect all explosive operations from effects of electric current originating in equipment such as soldering irons or guns, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding, separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator or squib. For example, circuits may be inadvertently completed from a defective electric soldering iron through a grounded contact. All electric type primers, detonators or squibs provided with wire leads shall have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Removable short circuiting clips, or other devices should be employed during manufacturing operations involving items such as electric primers, detonators or squibs.

6.4.3 Where necessary, protect personnel and operations from electromagnetic energy. In certain work with very sensitive explosives or items, a shielded environment free from significant electromagnetic energy should be provided.

6.5 Operating facility and storage regulations.

6.5.1 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance, at a minimum, with the American Table of Distances (ATD); and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with Intra-plant Distances set forth in the ATD when outside of loading rooms. For Navy managed or contractor managed Navy owned explosives loading plants, the Armed Services Explosives Safety Board standards approved by the Department of Defense, the Secretary of the Navy and the Naval Air Systems Command, covering quantity-distance regulations for explosives, apply."

Page 15, delete Notice in its entirety.

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